Angular Mini Project

Project Submission

1. Title Page

- **Project Title**: *Restaurant Menu and Ordering System using Angular and TypeScript*

Submitted By:

- *Team Members- Arvind V*

  Neha S

  Sahasra T

  Tanushree R

- *Roll Number- 2463009*

  2463041

  2463048

  2463062

- *College- arvind.v@btech.christuniversity.in*

  s.neha@btech.christuniversity.in

  Sahasra.t@btech.christuniversity.in

  tanushree.r@btech.christuniversity.in

- **Course**: *Angular*
- **Instructor Name**: *Pallavi*
- **Institution**: *Christ University*
- **Date of Submission**: *23/01/2026*

2. Abstract

This project focuses on developing a **Restaurant Menu and Ordering System** using **Angular and TypeScript**. The application allows users to browse menu items, view detailed dish information, add items to a cart, and place orders through an interactive interface. Angular components, routing, services, and dependency injection are used to build a modular and scalable architecture. TypeScript models ensure structured data handling for menu items, customers, and orders. The project demonstrates real-world application development using modern web technologies with an emphasis on usability and maintainability.

3. Objectives

- To design and develop a Restaurant Menu and Ordering System using Angular and TypeScript.
- To implement a component-based architecture for modular and reusable UI development.
- To enable users to browse menu items, view details, and manage a shopping cart efficiently.
- To apply Angular routing for smooth navigation between menu, cart, and checkout pages.
- To use TypeScript models and interfaces for structured and type-safe data handling.
- To implement services and dependency injection for cart management and shared data.
- To enhance user experience using Angular directives, data binding, and event handling.
- To demonstrate a real-world web application using modern front-end frameworks.

4. Scope of the Project

- To develop a front-end restaurant menu and ordering system using Angular and TypeScript.
- To allow users to browse menu items and add items to a cart.
- To implement component-based architecture and routing for smooth navigation.
- To manage cart data using Angular services and data binding.
- To use static/mock data without backend or payment integration.

5. Tools & Technologies Used

Tool/Technology	Purpose
Angular(v17)	Front-end framework for building the application
TypeScript	Structured and type-safe programming
HTML5	Page structure and content layout
CSS3	Styling and responsive design
Node.js	Runtime environment for Angular
Angular CLI	Project setup and development support
Visual Studio Code	Code editing and development
Static JSON Data	Mock backend data

6. HTML Structure Overview

- Uses semantic HTML elements for better readability and accessibility.
- Main layout includes a navigation bar, menu section, cart section, and checkout section.
- Menu items are displayed using reusable card-like containers.
- Each menu card contains an image, item name, price, and "Add to Cart" button.
- Router outlet dynamically loads pages such as Menu, Cart, and Checkout.
- Forms are used in the checkout section for collecting customer details.

7. CSS Styling Strategy

- CSS is used to create a clean and modern restaurant-themed UI.
- Flexbox and Grid layouts are applied for proper alignment and responsiveness.
- Hover effects are added to menu cards and buttons for better user interaction.

- Consistent color palette and typography enhance visual appeal.
- Buttons and cards are styled with rounded corners and shadows.
- Layout adapts well across different screen sizes.

8. Angular Framework Usage

- Application is built using Angular's component-based architecture.
- Components such as menu-list, cart, order-form, and navbar are modular and reusable.
- Angular directives (*ngFor, *ngIf) dynamically render menu items and cart contents.
- Services are used to manage shared data like cart items and total price.
- Routing enables smooth navigation between Menu, Cart, and Checkout pages.
- Dependency Injection ensures efficient data flow between components.

9. Angular CLI usage

- Angular CLI is used to create and manage the project structure.
- Components, services, and routing files are generated using CLI commands.
- ng serve is used to run the application locally for testing.
- CLI helps in building, debugging, and maintaining the application efficiently.
- Automatic recompilation occurs on file changes during development.
- Simplifies project setup and improves development productivity.

10. Key Features

Feature	Description
Responsive Design	Adapts seamlessly to all screen sizes
Smooth Navigation	Fixed top nav with anchor links

Cart Management	Add, remove, and view items with automatic total calculation
Smooth Navigation	Angular routing enables seamless page transitions
Service-Based Data Handling	Shared cart data managed using Angular services

11. Challenges Faced & Solutions

Challenge	Solution
Understanding Angular Component Structure	Followed Angular's modular architecture and used standalone components for better clarity.
Routing and Navigation Issues	Properly configured Angular Router with correct paths and route definitions.
Cart Data Not Persisting Across Pages	Implemented a shared CartService to manage and store cart data centrally.

12. Outcome

• Successfully developed a functional Restaurant Menu and Ordering System using Angular and TypeScript.
• Implemented modular components for menu display, cart management, checkout, and navigation.
• Enabled users to browse menu items, add items to the cart, view total cost, and proceed to checkout.
• Applied Angular routing for smooth navigation between Menu, Cart, and Checkout pages.

13. Future Enhancements:

• Add user authentication for login, profile management, and order history.

• Enhance forms with advanced validations and payment integration.

- Use Angular Material components more extensively for a polished UI.
- Implement advanced filtering, sorting, and search using custom pipes.
- Improve error handling with HTTP interceptors and user-friendly alerts.

14. Sample Code

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { RouterModule } from '@angular/router';
import { CartService, CartItem } from '../../services/cart';

@Component({
  selector: 'app-cart',
  standalone: true,
  imports: [CommonModule, RouterModule],
  templateUrl: './cart.html',
  styleUrls: ['./cart.css']
})
export class CartComponent {

  items: CartItem[] = [];
  total = 0;

  constructor(private cartService: CartService) {
    this.loadCart();
  }

  loadCart() {
    this.items = this.cartService.getItems();
```

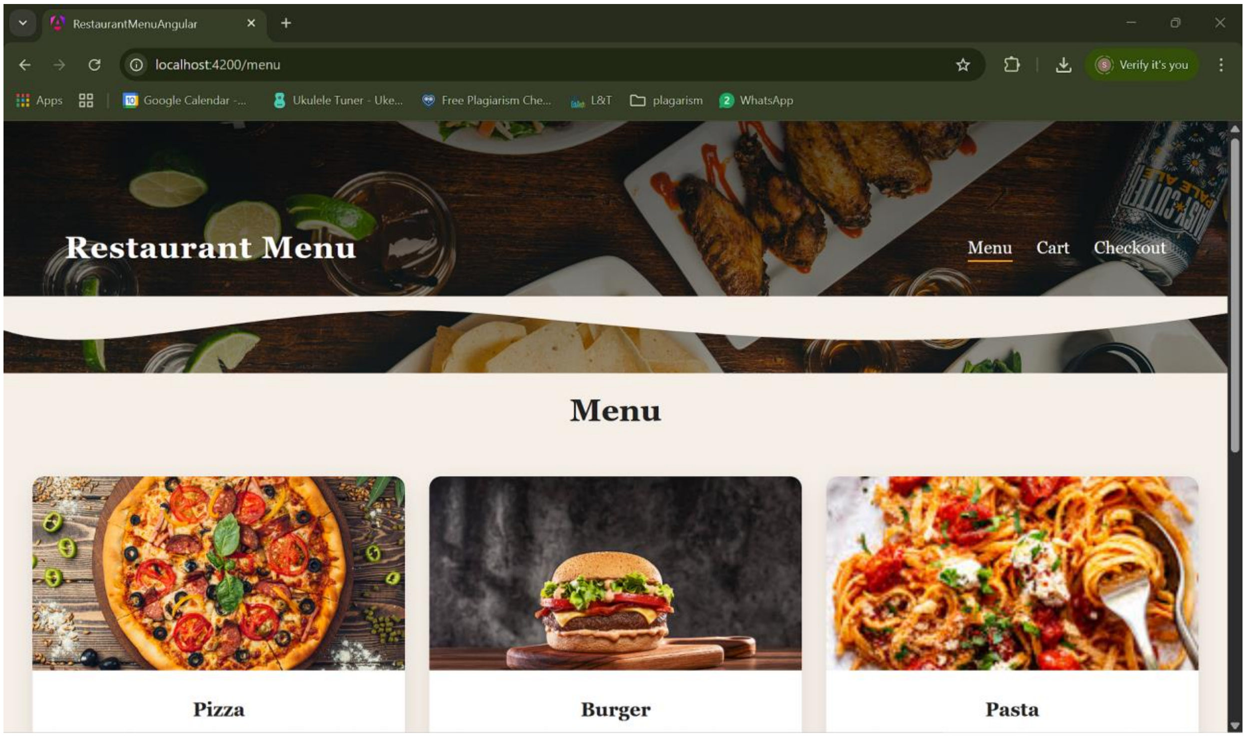
```
    this.total = this.cartService.getTotal();
  }

  removeItem(index: number) {
    this.cartService.removeItem(index);
    this.loadCart();
  }
}
```

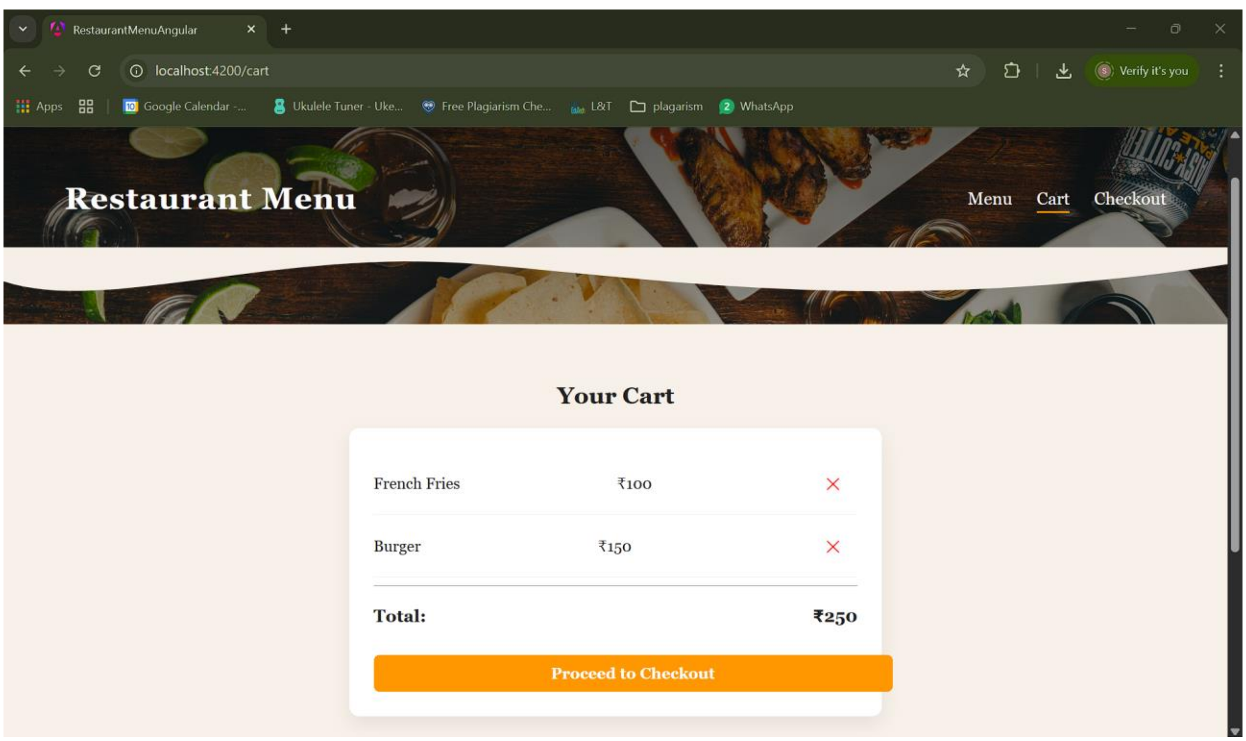
App.routes.ts

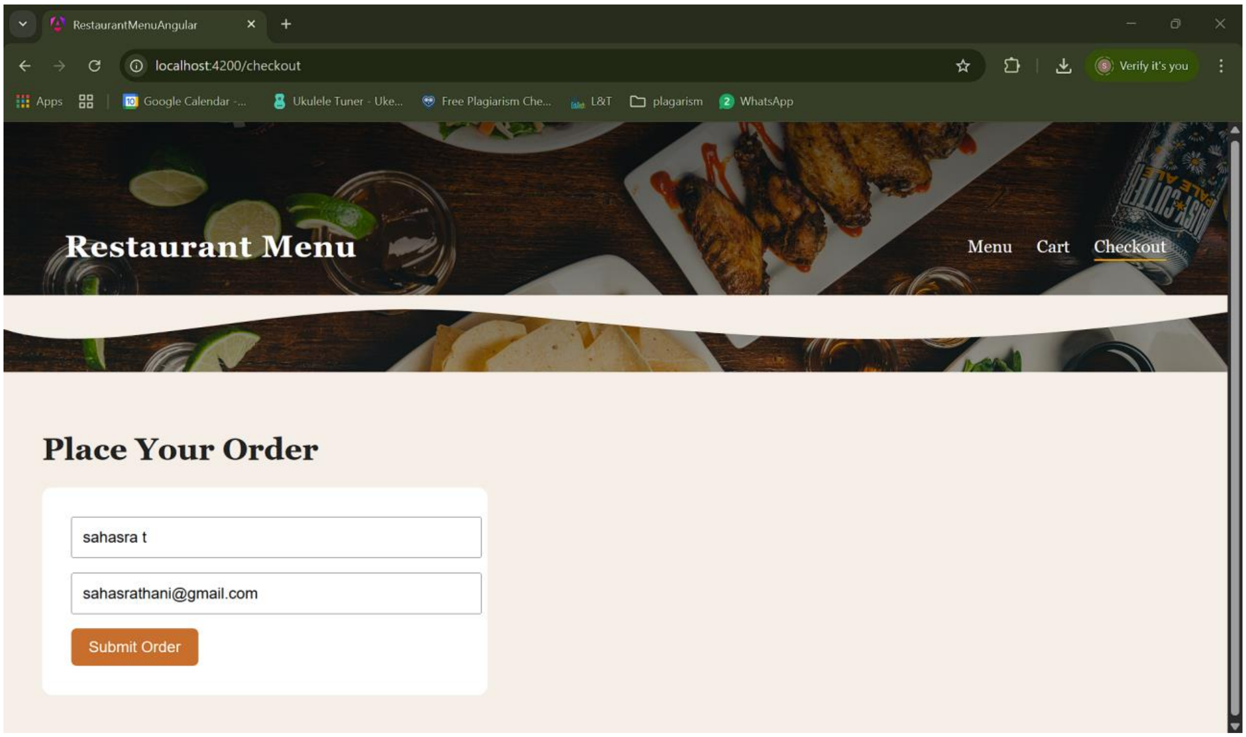
```
import { Routes } from '@angular/router';
import { MenuListComponent } from './components/menu-list/menu-list';
import { CartComponent } from './components/cart/cart';
import { OrderFormComponent } from './components/order-form/order-form';

export const routes: Routes = [
  { path: '', redirectTo: 'menu', pathMatch: 'full' },
  { path: 'menu', component: MenuListComponent },
  { path: 'cart', component: CartComponent },
  { path: 'checkout', component: OrderFormComponent }
];
```

15. Screenshots of Final Output

DESKT0P VIEW






RestaurantMenuAngular
localhost:4200/checkout
Verify it's you
Apps
Google Calendar -...
Ukulele Tuner - Uke...
Free Plagiarism Che...
L&T
plagarism
WhatsApp
Restaurant Menu
Menu
Cart
Checkout
Place Your Order
sahasra t
sahasrathani@gmail.com
Submit Order

16. Conclusion

The Restaurant Menu and Ordering System was successfully developed using Angular and TypeScript, demonstrating component-based architecture and efficient data handling. The application allows users to browse menu items, manage a cart, and place orders with smooth navigation. This project enhanced understanding of Angular concepts such as routing, services, and dependency injection. It provides a strong foundation for building scalable, real-world web applications.

17. References

- L&T LMS : https://learn.lntedutech.com/Landing/MyCourse